

# Spatial Data and QGIS

Turning maps into variables you can put in a regression

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Guest lecture · Empirical Development Economics · Undergraduate Economics, FEB UI

# Can you see the crisis?



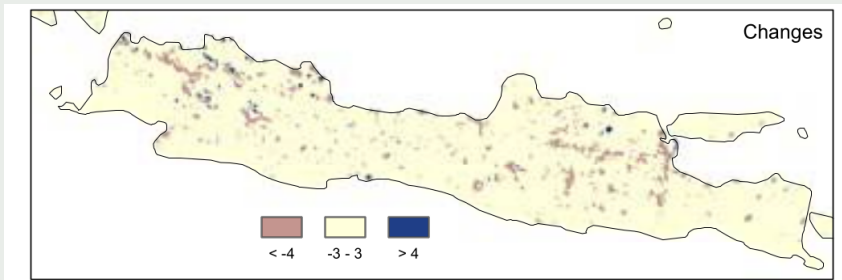
Figure: Henderson, Storeygard & Weil (2012), *AER* 102(2), Figure 3 (part).

Java, two years in a row. The 1998 crisis happened in between.

The change is small. Average brightness in 1998 is only about 3 percent lower, and city centres do not change at all because the sensor is already maxed out there.

Your eyes can barely catch it. That is exactly why we turn pictures into numbers.

## What your eyes miss



The third panel of the same figure: the *change* in light from 1997 to 1998, concentrated along the north coast city corridor.

**Think about it:** the authors did not trust their eyes either. They computed the difference first, then mapped it.

Figure: Henderson, Storeygard & Weil (2012), *AER* 102(2), Figure 3 (part).

# Open these two before we start

## The interactive lab

[billytulungen.com/lab](http://billytulungen.com/lab)

Five things you can click while we talk. Runs in your browser, nothing to install.

Raster resolution · Projections and buffers · Map classification · Joining tables · Units of analysis

## The practice files

[billytulungen.com/lab](http://billytulungen.com/lab) → download

Twelve layers and tables, 430 KB. Unzip anywhere and open them straight from QGIS.

The README inside says which file each exercise needs. Everything is synthetic data built for this class.

## Takeaway

You also need QGIS 3.x, the LTR release, from [qgis.org](http://qgis.org). Please install it before the session starts.

# Why Location Matters

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# Development questions always have a location

## Infrastructure

Does a new road reduce poverty in a village?

You need the distance from every village to the road, year by year.

## Natural resources

Do new plantations raise local wages?

You need yearly land cover change at the sub-district level.

## Climate risk

How much rice output is lost to drought?

You need rainfall by district and by growing season.

## Takeaway

All three have one thing in common: you have to build the key variable yourself, from a map. It is not published in any statistics book.

# The first law of geography

*“Everything is related to everything else, but near things are more related than distant things.”*

Waldo Tobler, 1970

What this means for econometrics: **observations that sit close together are not independent**. We come back to this at the end.

# From light to an economic variable



FIGURE 3. ASIAN FINANCIAL CRISIS: JAVA, INDONESIA

Dashed line: the GDP that was recorded. Solid line: GDP predicted from night lights alone.  
Both drop together in 1998.

**Think about it:** Which kinds of places would a proxy like this be most useful for?

Figure: Henderson, Storeygard & Weil (2012), *AER* 102(2), Figure 3 (part).

# Dots that became data: transmigration

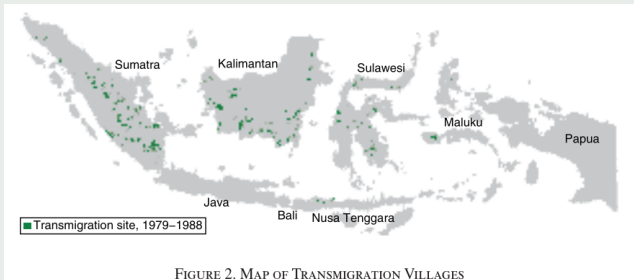


FIGURE 2. MAP OF TRANSMIGRATION VILLAGES

Each green dot is one transmigration village, 1979–1988.

The authors turned a 1998 government census into a layer of points, then matched it with climate and soil grids.

This is exactly the workflow you will learn today, just with real data and a bigger question.

Figure: Bazzi, Gaduh, Rothenberg & Wong (2016), *AER* 106(9), Figure 2.

# Forced cultivation in Java: three data types in one map

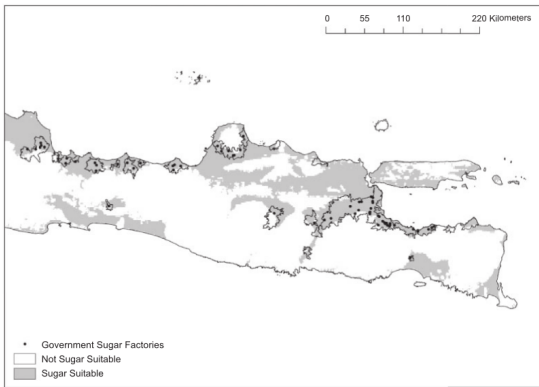


FIGURE 3  
The Cultivation System.

Sources: [Commissie Umbgrove \(1858\)](#). Black dots are government sugar factories. Lines are forced cultivation catchment area boundaries. Gray areas have an elevation and slope that are suitable for sugar cultivation, whereas white areas do not.

Black dots: colonial sugar factories.  
Lines: the boundary of each forced cultivation area. Grey: land suitable for sugar.

Points, lines and a grid – all three in one picture, and all three are things you will build yourself today.

Those boundaries are what the authors use to measure the long-run effect.

Figure: Dell & Olken (2020), *Review of Economic Studies* 87(1), Figure 3.

## Market access: a network becomes one index



Donaldson and Hornbeck combine the railway network with the size of every neighbouring market into **one market access number per county**, then map how it changed from 1870 to 1890.

The original caption says the counties are split into seven groups with *the same number of counties* in each group.

That is **quantile** classification – the same choice we compare later today.

Figure: Donaldson & Hornbeck (2016), *QJE* 131(2), Figure III.

# Spatial data in economics research

| Study                                | Spatial data used                       | What it was for                                 |
|--------------------------------------|-----------------------------------------|-------------------------------------------------|
| Henderson, Storeygard & Weil (2012)* | Satellite night lights                  | A stand-in for growth where statistics are weak |
| Dell (2010)*                         | Historic <i>mita</i> boundary, Peru     | Geographic discontinuity for identification     |
| Donaldson (2018)                     | Colonial railway network, India         | Trade costs between regions                     |
| Donaldson & Hornbeck (2016)*         | US railways                             | Building a market access index                  |
| Bazzi et al. (2016)*                 | Transmigration origins and destinations | Skill transferability in Indonesia              |
| Dell & Olken (2020)*                 | Forced cultivation boundaries, Java     | Long-run effects of a colonial economy          |
| Donaldson & Storeygard (2016)*       | Satellite imagery across fields         | A survey of satellite data in economics         |
| Harari & Wong (2026)*                | Slum upgrading boundaries, Jakarta      | Long-run effects of upgrading policy            |

\* The original figure is shown on a nearby slide. **The point:** none of them started from a map. They started from an economic question, and used maps because there was no other way to measure what they needed.

# What spatial data is actually for

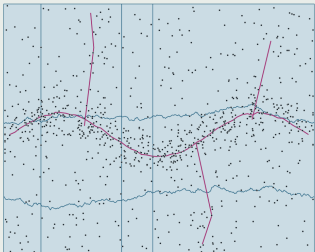
- 1 **It creates new variables.** Images and maps become columns of numbers: distance, area, elevation, brightness, rainfall.
- 2 **It creates identification.** Borders, coastlines and transport networks give you variation you can argue is exogenous.
- 3 **It helps you check your work.** Mapping your residuals or your treatment shows patterns a regression table hides.

# Spatial Data Basics

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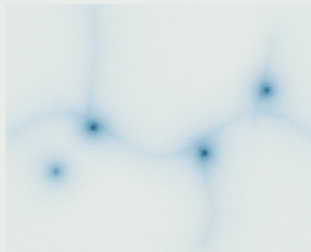
# Shapes with edges, or a grid of numbers

Vector



District polygons, roads and village points: three separate layers, each with its own table.

Raster



Night lights on a grid of cells. One number per cell, nothing else.

## Vector: points, lines and polygons

A vector file holds coordinates and a table, nothing else. Which of the three shapes it holds decides what you can do with it.

| Shape    | Typical layers                 | What you build from it              |
|----------|--------------------------------|-------------------------------------|
| Points   | Schools, clinics, markets      | Distance, density, service coverage |
| Lines    | Roads, railways, rivers, power | Transport cost, connectivity        |
| Polygons | Provinces down to villages     | Usually the rows of your dataset    |

One file holds one shape only. District polygons and school points always arrive as separate layers, and you join them yourself.

# Raster: the world as a grid of numbers



Resolution is how wide one cell is on the ground. A raster has no attribute table: to use it in a regression, you must summarise its values inside a boundary. That step is called *zonal statistics*.

| Resolution | Example data                 |
|------------|------------------------------|
| 30 m       | Forest cover, SRTM elevation |
| ~500 m     | VIIRS night lights           |
| ~5 km      | CHIRPS rainfall              |

Figure: Henderson, Storeygard & Weil (2012), *AER* 102(2), Figure 1.

## File formats you will meet

| Format               | Extension           | What to remember                                     |
|----------------------|---------------------|------------------------------------------------------|
| Shapefile            | .shp .shx .dbf .prj | Move all the files, not just the .shp                |
| GeoPackage           | .gpkg               | One file, many layers. QGIS prefers this             |
| GeoJSON              | .geojson            | Plain text, easy to inspect                          |
| GeoTIFF              | .tif                | Raster. Stores its coordinate system inside          |
| CSV with coordinates | .csv                | Points only. Needs a latitude and a longitude column |

### Watch out

The .prj file holds the coordinate system. Without it, QGIS does not know which part of the planet your map belongs to.

# The attribute table: your bridge to Stata and R



|   | fid | kode_kab | nama_kab        | reformasi_izin | luas_km2 | n_desa |
|---|-----|----------|-----------------|----------------|----------|--------|
| 1 | 1   | 6201     | Sungai Hulu     | 1              | 6085,7   |        |
| 2 | 2   | 6202     | Batang Tengah   | 0              | 12584,3  |        |
| 3 | 3   | 6203     | Muara Baru      | 1              | 5564,8   |        |
| 4 | 4   | 6204     | Tanjung Lestari | 0              | 31913    |        |
| 5 | 5   | 6205     | Bukit Raya      | 1              | 9484,9   |        |
| 6 | 6   | 6206     | Danau Panjang   | 0              | 22271,6  | 1      |
| 7 | 7   | 6207     | Kuala Sepakat   | 1              | 7729     |        |
| 8 | 8   | 6208     | Rimba Jaya      | 0              | 39730,2  | 2      |

Behind every vector layer sits an ordinary table. One row is one shape on the map.

The code column is your join key. Area is computed for you.

Once geometry becomes columns of numbers, the rest is econometrics you already know.

# Coordinate systems and projections

The Earth is curved and your screen is flat, so every layer carries one rule for translating between the two. Three things to keep straight:

|                | Example                                       | What it is for                                                      |
|----------------|-----------------------------------------------|---------------------------------------------------------------------|
| Geographic CRS | <a href="#">EPSG:4326</a>                     | Latitude and longitude in degrees. Storing and sharing              |
| Projected CRS  | <a href="#">EPSG:327xx</a>                    | Units in metres. Required before any distance, area or buffer       |
| Datum          | <a href="#">WGS 84</a> / <a href="#">SRGI</a> | The reference shape of the Earth. WGS 84 is fine for most economics |

## Watch out

QGIS reprojects layers so they *look* right together, but it computes distance and area using each layer's own CRS. A correct picture does not mean correct numbers.

## Choosing a CRS for Indonesia

| What you are doing                       | System            | EPSG        |
|------------------------------------------|-------------------|-------------|
| Storing and sharing data                 | WGS 84 geographic | 4326        |
| Distance and area in Sumatra             | UTM zone 46–47    | 32746–32747 |
| Distance and area in Java and Kalimantan | UTM zone 48–49    | 32748–32749 |
| Bali, Nusa Tenggara, Sulawesi            | UTM zone 50–51    | 32750–32751 |
| Maluku and Papua                         | UTM zone 52–54    | 32752–32754 |

Codes starting 327xx are southern zones, 326xx are northern ones. Indonesia spans nine UTM zones, so for a national analysis either work zone by zone or use an equal-area projection.

## Units of analysis and BPS codes

| Level           | How many     | Code digits | Note                                              |
|-----------------|--------------|-------------|---------------------------------------------------|
| Province        | 38           | 2           | Too coarse for most micro questions               |
| District / city | $\pm 514$    | 4           | The most common unit in published statistics      |
| Sub-district    | $\pm 7,200$  | 7           | Fine enough for local variation, still manageable |
| Village         | $\pm 84,000$ | 10          | Available in the village census                   |

### Watch out

When districts split, their codes change from year to year. For a long panel, build a crosswalk table or map every year back to the boundaries of one base year.

# When the unit of analysis changes the answer

## Scale effect

Take the same micro data and average it up to provinces, then districts, then sub-districts. You get different coefficients. Sometimes different signs.

## Zoning effect

Even with the same number of units, where you draw the lines changes the result. Administrative borders are not economic borders.

Choose your unit from the **economic mechanism you believe in** – a local labour market, a river basin, a commuting zone – not from whichever file was easiest to download.

Then report the result at another unit as a robustness check. A referee will ask.

Related trap: a relationship that holds across districts does not have to hold across the people inside them. The name for all of this is the Modifiable Areal Unit Problem (Openshaw 1984).

# Where to get spatial data for Indonesia

| Source                    | What it has                    | Notes                                                   |
|---------------------------|--------------------------------|---------------------------------------------------------|
| BPS statistical area maps | Provinces down to villages     | Codes match published statistics. Start here            |
| Ina-Geoportal (BIG)       | Base maps and official borders | The official reference if borders are disputed          |
| GADM                      | Global administrative borders  | Quick for practice; not always up to date               |
| OpenStreetMap             | Roads, buildings, facilities   | Coverage varies a lot between regions                   |
| VIIRS and DMSP            | Night lights                   | A proxy for activity; be careful joining the two series |
| WorldPop                  | Population density at 100 m    | A denominator for per-capita variables                  |
| CHIRPS                    | Daily rainfall since 1981      | The usual source of weather shocks                      |
| Global Forest Change      | Forest cover at 30 m           | Yearly deforestation down to the pixel                  |
| SRTM / DEM                | Elevation and slope            | Geographic controls and instruments                     |

# The data behind these slides

## What the exercises run on

District polygons, village points, a road network, a night lights raster, and one statistics table.

These files are **synthetic**. They were generated for this class so the patterns read clearly, and every place name in them is invented. Nothing here describes a real district.

## Takeaway

Practising on synthetic data is fine. Reporting results from it is not. Swap in the real sources before anything goes near a thesis.

## Where the real versions live

Each layer has a real counterpart on the sources slide: BPS for boundaries and statistics, BIG for official base maps, OpenStreetMap for roads, VIIRS for night lights, CHIRPS for rainfall.

## Figures from published research

Every borrowed figure carries its full citation on the slide where it appears.

# Spatial Operations

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## The operations you keep coming back to

| Operation          | What it does                                                                                      |
|--------------------|---------------------------------------------------------------------------------------------------|
| Attribute join     | Attaches a statistics table to polygons through a shared code. Exactly like <i>merge</i> in Stata |
| Spatial join       | Attaches attributes by location: which point falls inside which polygon                           |
| Buffer             | Draws a zone of fixed distance around points, lines or areas                                      |
| Clip and intersect | Cuts one layer using the boundary of another                                                      |
| Dissolve           | Merges small polygons into larger units                                                           |
| Zonal statistics   | Reduces millions of raster cells to one number per area                                           |

Almost every variable in an applied paper is some combination of these six, run in the right coordinate system and exported to a table.

# Joining by code, joining by location

## Attribute join

The key is a column both tables share, usually the district code.

```
polygon.code = table.code
```

Geometry is not used at all.

## Spatial join

The key is the geometry itself. No shared column needed.

```
intersects | within | contains |  
nearest
```

Use this when your data only has coordinates.

## Takeaway

An example that uses both: how many clinics are within 5 km of each village? Buffer first, then spatial join with *intersects*, then count.

## Zonal statistics: a raster becomes rows of data



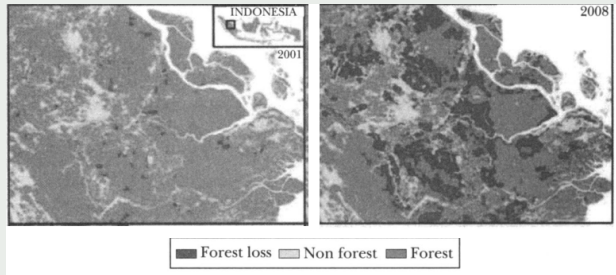
The magenta line marks one zone.  
The tool reads every cell inside it and  
returns a single number for that  
district.

### One district

|       |                        |
|-------|------------------------|
| count | 5,909 cells            |
| mean  | 3.299                  |
| sum   | 19,492.7               |
| area  | 29,096 km <sup>2</sup> |

Always ask for **count** too: a small district  
may contain only a dozen cells.

## Repeat it every year: deforestation in Riau



The same patch of Riau province in 2001 and in 2008. A yearly raster summarised by area, then stacked into a district-by-year panel.

**Think about it:** with eight years like this, what variable would you build?

Figure: Donaldson & Storeygard (2016), *JEP* 30(4), Figure 1 (two of eight panels); original source Burgess et al. (2012).

# The standard workflow

1. **Collect your layers.** Keep the raw files in their own folder and never edit them.
2. **Put everything in one CRS.** Reproject to a metre-based system before anything involving distance or area.
3. **Build your variables.** Buffer, join, dissolve, zonal statistics. Each step produces one column with an economic meaning.
4. **Export the attribute table** to CSV, with the area code and the year included.
5. **Estimate in Stata or R.** From here on you do not need QGIS again.

## Takeaway

Record steps 2 to 4 in the *Graphical Modeler*. A saved workflow can be re-run for the next year, and you can attach it as proof that your work replicates.

# Hands-on QGIS

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# Why QGIS

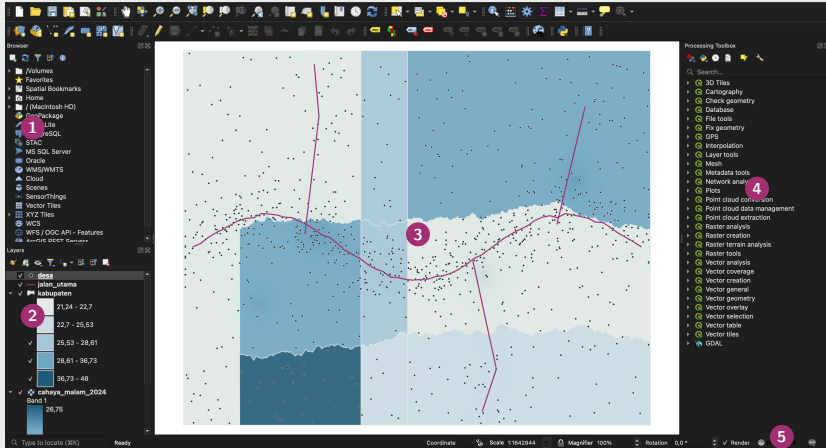
There is no licence to expire when you graduate or move institution, which also means anyone can rerun your work without buying anything.

Almost everything applied economics needs is in the standard install: joins, buffers, zonal statistics, thematic maps. Where it stops, it hands over to GDAL, GRASS, SAGA or its own Python console rather than blocking you.

The part that matters most for a thesis is quieter. The Processing Toolbox logs every step you run, and the Graphical Modeler turns a sequence of steps into a saved workflow you can run again next year, or attach as evidence that your results replicate.

Download the **LTR** release from [qgis.org](https://qgis.org). The standard install is enough for today. No extra plugins.

# The QGIS window, and the parts you actually use



# What each part is for

1. **Browser panel.** Browse files on your computer, databases and online services without leaving QGIS.
2. **Layers panel.** The layers you have open. Top to bottom is drawing order, and the legend for each class appears here.
3. **Map canvas.** Your working view, not the final product. Maps for a thesis are made in Print Layout.
4. **Processing Toolbox.** Every analysis tool. You do not need to memorise menus: just type the tool name in the search box.
5. **Status bar.** Scale, cursor coordinates, and the project EPSG code in the bottom right corner.

Lost a panel? `View > Panels`    `Toolbox: Ctrl+Alt+T`

# Which file for which exercise

From [billytulungen.com/lab](http://billytulungen.com/lab) → download, 430 KB. The same list sits in the README inside the zip.

| Exercise                      | Open these                                                                       |
|-------------------------------|----------------------------------------------------------------------------------|
| 1 Open a layer, check its CRS | <a href="#">kabupaten.gpkg</a>                                                   |
| 2 Join a statistics table     | <a href="#">kabupaten_polos.gpkg</a> and <a href="#">statistik_kabupaten.csv</a> |
| 3 Thematic map                | <a href="#">kabupaten.gpkg</a>                                                   |
| 4 Point layer from a CSV      | <a href="#">village_coordinates.csv</a>                                          |
| 5 Buffer and spatial join     | <a href="#">desa.gpkg</a> and <a href="#">jalan.gpkg</a>                         |
| 6 Zonal statistics            | <a href="#">nightlights.asc</a> and <a href="#">kabupaten.gpkg</a>               |
| 7 Print Layout                | the map you built in Exercise 3                                                  |

## Watch out

Everything in the bundle is stored in [EPSG:4326](#), which is degrees. Reproject to [EPSG:32749](#) before Exercise 5, or any step that measures distance or area.

# Exercise 1: open a layer and check its CRS

1. Layer > Add Layer > Add Vector Layer, pick `kabupaten.gpkg`.
2. Right-click the layer, **Properties** > **Information**, and read the CRS line.
3. Compare it with the EPSG code in the bottom right. That is the *project* CRS, not the layer CRS.
4. Right-click the layer, **Open Attribute Table** (F6). Count the rows.
5. Use **Identify Features**, click one polygon, and read its attributes.

Open first: `kabupaten.gpkg`



The screenshot shows the QGIS interface with the 'Properties' dialog box open for the 'kabupaten.gpkg' layer. The 'Information' tab is selected, displaying metadata about the layer's storage and geometry. The 'Coordinate Reference System (CRS)' tab is also visible, showing the layer's CRS as EPSG:4326 - WGS 84, which is Geographic (uses latitude and longitude for coordinates).

| Information from provider |                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------|
| Storage                   | GPKG                                                                                 |
| Encoding                  | UTF-8                                                                                |
| Geometry                  | Polygon (Polygon)                                                                    |
| Extent                    | 111.0000000000000000, -3.5000000000000000 : 116.0000000000000000, 0.5000000000000000 |
| Feature count             | 12                                                                                   |

| Coordinate Reference System (CRS) |                                                                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Name                              | EPSG:4326 - WGS 84                                                                                                 |
| Units                             | Geographic (uses latitude and longitude for coordinates)                                                           |
| Type                              | Geographic (2D)                                                                                                    |
| Method                            | Lat/long (Geodetic alias)                                                                                          |
| Celestial Body                    | Earth                                                                                                              |
| Accuracy                          | Based on <i>World Geodetic System 1984 ensemble</i> (EPSG:6326), which has a limited accuracy of at best 2 meters. |
| Reference                         | Dynamic (relies on a datum which is not plate-fixed)                                                               |

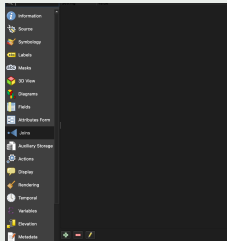
## Watch out

The CRS block names the units. *Geographic* means degrees, which cannot measure distance or area. Reproject first.

## Exercise 2: join a statistics table to the map

1. The key is `kode_kab` on the map but `kode` in the CSV.
2. **Layer > Add Layer > Add Delimited Text Layer**, choose *No geometry*.
3. Right-click the polygon layer, **Properties > Joins**, press plus, and set the key field on each side.
4. Tick *Custom field name prefix* so column names do not clash.
5. Open the attribute table and check that none of the new column is empty.

Open first: `kabupaten_polos.gpkg` and `statistik_kabupaten.csv`



### Takeaway

The plus button sits below the empty join list. Nothing in this panel ever reports a failed join; check the attribute table yourself.

# How a join fails without telling you

| Symptom                      | Cause                                                   | Fix                                               |
|------------------------------|---------------------------------------------------------|---------------------------------------------------|
| Every joined column is empty | Key types differ: text on one side, number on the other | Make them match with the Field Calculator         |
| Some rows are empty          | Codes changed when districts split                      | Use a crosswalk table between years               |
| Leading zeros disappear      | Excel read the code column as a number                  | Import the CSV with the key column forced to text |
| Row count grows              | The key is not unique on one side                       | Check for duplicates before joining               |
| Names do not match           | Spelling, abbreviations and capitalisation differ       | Join on codes, never on names                     |

## Watch out

A partly failed join produces no error message. The regression still runs, and the sample is no longer random. Count rows before and after.

## Exercise 3: make a thematic map

1. Right-click the layer, **Properties > Symbology**.
2. Change *Single Symbol* to *Graduated*.
3. Set *Value* to `kemiskinan`, the poverty rate.
4. Pick a *Color ramp* and a *Mode*, then press *Classify*.
5. Set the number of classes. Five is enough for almost any map.
6. Apply, then change *Mode* and watch what happens.

Open first: `kabupaten.gpkg`



### Takeaway

Step 6 is the real lesson: the same map can tell different stories.

# One column of numbers, three different maps

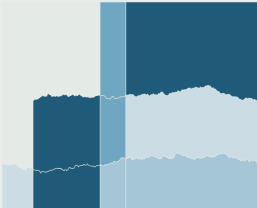
Equal Interval



26.6 · 31.9 · 37.3 · 42.7

Cuts spread evenly across the range.

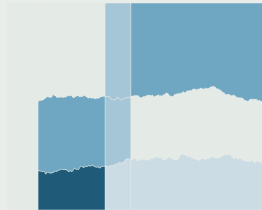
Quantile



22.2 · 23.8 · 26.4 · 30.9

All four cuts sit below 31.

Natural Breaks



22.7 · 25.5 · 28.6 · 36.7

Cuts fall where the data is thin.

The same column, the same range of 21.2 to 48.0, five classes, the same colour ramp. Only the mode changed.

**Think about it:** which map would you pick if you wanted poverty to look evenly spread?

# Choosing a classification and a colour scheme

| Mode               | When to use it                     |
|--------------------|------------------------------------|
| Equal Interval     | Data spread fairly evenly          |
| Quantile           | Same number of areas in each class |
| Natural Breaks     | A safe default for skewed data     |
| Standard Deviation | To highlight who deviates          |

**Sequential** for a quantity that increases.

**Diverging** for values with a meaningful midpoint, such as positive and negative growth.

**Avoid rainbow schemes.** They have no natural order, and readers with red-green colour blindness cannot read them.

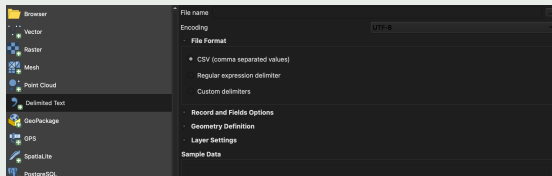
## Watch out

If your variable is a count, divide it by population or by area first. Otherwise your map just shows where people live, and Java will always be dark.

## Exercise 4: make a point layer from a CSV

1. The columns are named **X** (longitude) and **Y** (latitude).
2. **Layer > Add Layer > Add Delimited Text Layer.**
3. Choose *Point coordinates*. Set **X = longitude** and **Y = latitude**.
4. Set *Geometry CRS* to EPSG:4326 if your coordinates are in degrees.
5. Check the points land in Indonesia, then **Export > Save Features As.**

Open first: `village_coordinates.csv`



### Watch out

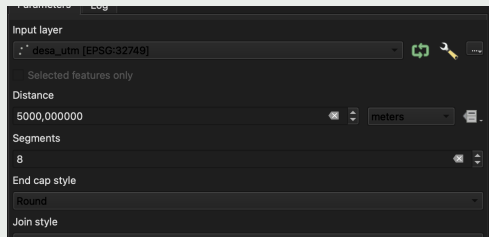
X and Y live inside *Geometry Definition*, which opens once a file is chosen. X is longitude, Y is latitude.

## Exercise 5: buffer and spatial join

**The question:** which villages sit within 5 km of a main road?

1. Reproject both to `EPSG:32749` with `Reproject Layer`.
2. `Vector > Geoprocessing Tools > Buffer` on the roads, *Distance* 5000.
3. `Vector > Data Management > Join Attributes by Location`.
4. Villages as the base layer, the buffer as the join layer, predicate *intersects*.
5. Check your answer against `jarak_jalan_km_TRUE`, the real distance.

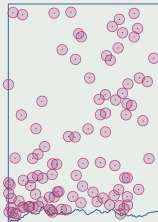
Open first: `desa.gpkg` and `jalan.gpkg`



The layer is in EPSG:32749, so the unit box beside *Distance* reads *meters*. Step 1 is what puts it there.

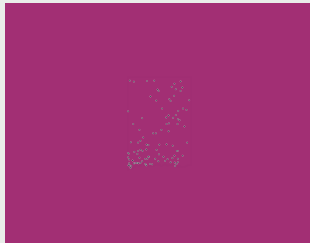
# Typing 5000 in the wrong CRS

Layer in metres EPSG:32749



A 5,000 metre buffer. The circles make sense.

Layer in degrees EPSG:4326



The same number read as 5 degrees, about 555 km.

## Watch out

Both of these came out of QGIS without a single error message. The one on the right still ran, still produced a new column, and could still end up in your regression.

## The same buffer, continental scale

A. 40-Mile Buffers: Waterways (Black) and Railroads (Gray)



The operation you just ran in Exercise 5 is the same operation used in the *Quarterly Journal of Economics*.

Forty-mile buffers around railways (grey) and waterways (black) in 1890. Panel B of the original adds canals that were once proposed, as a comparison.

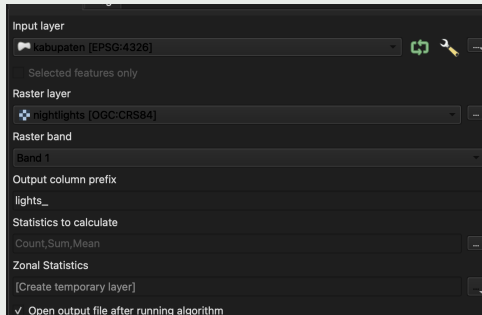
The tool is not what makes it research.  
The question is.

Figure: Donaldson & Hornbeck (2016), *QJE* 131(2), Figure I.

## Exercise 6: zonal statistics and export to CSV

1. Layer > Add Layer > Add Raster Layer, pick `nightlights.asc`.
2. Open the Processing Toolbox and type `Zonal statistics`.
3. Set the zones to the district polygons and the raster to the lights layer.
4. Tick the statistics you need: *mean*, *sum*, *count*.
5. Run it, then **Export > Save Features As** in CSV format.
6. Repeat for the next year and stack them into a panel.

Open first: `nightlights.asc` and `kabupaten.gpkg`



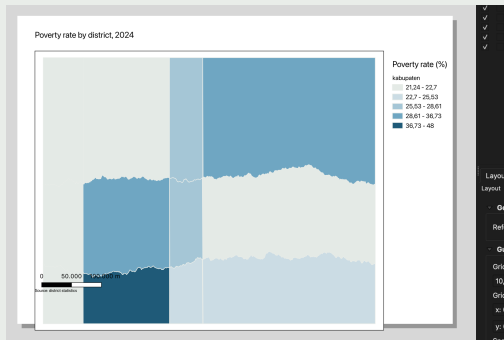
### Watch out

Read the two CRS labels here. A mismatch returns an empty result.

## Exercise 7: Print Layout for your thesis map

1. **Project > New Print Layout**, name it after the figure number.
2. **Add Item > Add Map** and drag a box across the page.
3. Add a *Legend*, a *Scale Bar* and a north arrow.
4. Use *Add Label* for the title and the data source line.
5. Rename the class labels in the legend by double-clicking them.
6. **Layout > Export as PDF**, or as an image at 300 dpi.

Open first: the map you built in Exercise 3



### Takeaway

Every element appears in the Items panel on the right. A screenshot of the map canvas gives you none of this.

# Checklist before a map leaves your desk

- The title says what, where and when.
- The legend uses clear units, not raw column names.
- There is a scale bar and a north arrow.
- The data source and year are given.
- Counts have been turned into rates.
- The classification method and number of classes are stated.
- The colours still work when printed in black and white.
- Areas with no data look different from areas with a value of zero.

The last one is missed most often, and it changes what readers conclude more often than any of the others.

## Back to Empirical Work

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## From QGIS output to a regression

$$y_{it} = \alpha + \beta \text{road\_distance}_i + \gamma \text{lights}_{it} + \delta \text{rain}_{it} + \mu_i + \lambda_t + \varepsilon_{it}$$

| Variable      | Built with                        | From which layer                       |
|---------------|-----------------------------------|----------------------------------------|
| road_distance | Distance to Nearest Hub           | Road network and village centre points |
| lights        | Zonal statistics, mean            | Night lights raster, one per year      |
| rain          | Zonal statistics, then aggregated | Daily rainfall raster                  |
| area          | Field Calculator, area expression | Polygons in a metre-based CRS          |

### Takeaway

You should be able to explain every row of this table in your defence. *Where did this number come from* is the first question an examiner asks.

# Geographic discontinuity: the mita boundary in Peru

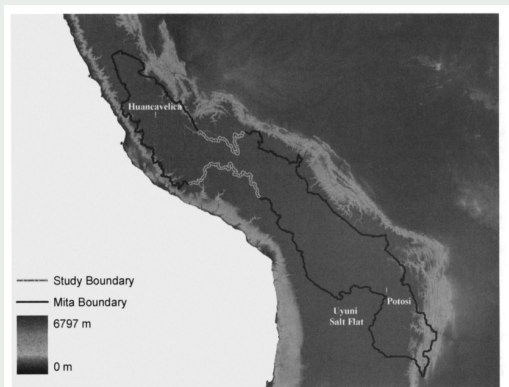


FIGURE 1.—The *mita* boundary is in black and the study boundary in light gray. Districts falling inside the contiguous area formed by the *mita* boundary contributed to the *mita*. Elevation is shown in the background.

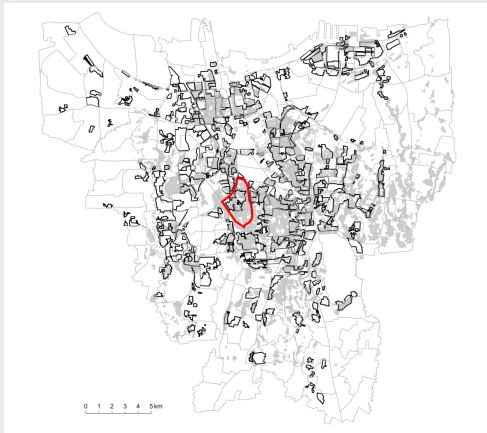
The black line is the boundary of a forced mining labour system that ran from 1573 to 1812. The background is an elevation raster.

Villages on either side of this line have similar geography, ethnicity and tax history. What differs is whether they were subject to the system.

An administrative border is treated as a treatment line – and the whole identification strategy rests on one vector layer.

Figure: Dell (2010), *Econometrica* 78(6), Figure 1.

## The same idea in Jakarta



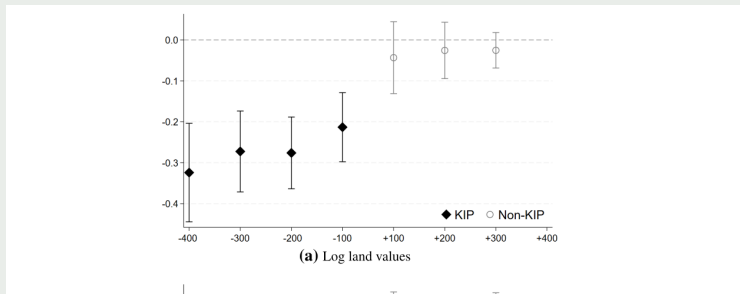
Black outlines: areas upgraded by the KIP programme between 1969 and 1984. Shaded grey: kampungs that already existed before it. The red shape in the middle is the central business district.

The programme reached 5 million people and covered a quarter of Jakarta's land.

Someone had to digitise those boundaries first. That layer of polygons is what makes the whole study possible.

Figure: Harari & Wong (2026), *Review of Economic Studies* 93(4), Figure 1.

# What a distance variable looks like in the end



Land values in 100 metre bands either side of a KIP boundary. Negative is inside the upgraded area.

**Think about it:** that horizontal axis is a distance column computed in a GIS. This is where Exercise 5 ends up.

Figure: Harari & Wong (2026), *Review of Economic Studies* 93(4), Figure 3, panel (a).

# Space as a source of identification

| Strategy                  | What it rests on                                                                                                  |
|---------------------------|-------------------------------------------------------------------------------------------------------------------|
| Geographic discontinuity  | Places either side of a policy border, assuming everything else changes smoothly across it                        |
| Distance as an instrument | Distance to old infrastructure or to a resource, if you can argue it reaches the outcome through one channel only |
| Market access index       | A transport network and the size of neighbouring markets, combined into one measure of connectedness              |
| Weather shocks            | Rainfall and temperature by area and season, variation that is hard to argue people chose                         |

## Watch out

Being close together does not make something exogenous. Administrative borders are often drawn *because* of a difference: a river, a mountain, an ethnic line, or local politics.

## Neighbours share shocks, so fix your standard errors

Weather, labour markets and local policy do not stop at a district border. Residuals end up correlated across space, and ordinary OLS standard errors come out too small.

1. Build a spatial weights matrix, either by adjacency or by a distance cut-off.
2. Test Moran's I on the residuals. GeoDa does it, so does the R package `spdep`.
3. If it fails, use Conley (1999) standard errors, a spatial HAC correction that lets correlation fade with distance. Stata has `acreg` and `ols_spatial_HAC`; R has `conleyreg`.

A simpler route is to cluster at a larger unit, so long as you still have enough clusters to make that honest.

## Where it usually goes wrong

1. **Mixing coordinate systems.** Distances and areas become meaningless, and there is almost never an error message.
2. **A partly failed join.** Some areas drop out of the sample, the regression still runs, and the result is biased.
3. **Mapping counts instead of rates.** What you get is a population map with a different title.
4. **Reading area-level patterns as individual ones.** A relationship across districts need not hold across the people in them.
5. **A beautiful map with no question.** A map supports an argument. It is not the argument.

## Practice on your own

Take district boundaries and one development indicator of your choice.

1. Join the table to the polygons by area code. Report how many rows failed to match, and why.
2. Make a five-class thematic map, and explain your choice of classification.
3. Compute the distance from each district capital to the nearest port.
4. Export the table to CSV and regress your indicator on that distance.
5. Hand in one page: map, regression table, three sentences of interpretation.

### Takeaway

What is assessed is not the size of the coefficient, but whether you can explain where every number came from and see the weaknesses yourself.

# Where to learn more

## Learning QGIS

QGIS Training Manual  
[docs.qgis.org](https://docs.qgis.org)

QGIS Tutorials and Tips  
[qgistutorials.com](https://qgistutorials.com)

## Moving to code

Geocomputation with R  
[r.geocompx.org](https://r.geocompx.org)

Spatial Data Science  
[r-spatial.org/book](https://r-spatial.org/book)

## Finding data

Ina-Geoportal, BIG  
[tanahair.indonesia.go.id](https://tanahair.indonesia.go.id)

BPS statistical area maps  
[bps.go.id](https://bps.go.id)

For anything you repeat, write a script. Use QGIS to look, and R or Python to process.

A map is not the end of your research.

It is how you build a variable  
that did not exist before.

# Thank you

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